

MFE 230GA · FINAL PROJECT · FALL 2026

Follow the Workers: Results Book

Every result, figure, table and claim behind the report, with its source file

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COURSE

MFE 230GA – Active Equity Management

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Contents

1	Verdict	2
2	Scoreboard	2
3	Figures	2
4	Statistics	8
5	V2d robustness battery (v3)	12
6	The agent experiment	17
7	AI-interaction inventory	18
8	Claims register	20
9	Known issues	23

1 Verdict

VERDICT.md is written at Stage 3.

2 Scoreboard

IC is one-month except V2d (12-month, its holding horizon). DSR is the research-period value with 41 nominal trials for the frozen studies and the test-window value with 112 trials for v2. Labels: conf. = confirmatory, expl. = exploratory follow-up. Source: [results/scoreboard.csv](#).

Strategy	Label	Res.	Test	Full	18m	IC (t)	α/m (t)	Plac.	DSR	Verdict
A0 fixed rule (Sonnet)	conf.	-0.44	-0.36	-0.48	-1.56	-0.020 (-0.71)	-0.25% (-0.92)		0.00	negative, not primary
A1 ridge (Sonnet)	conf.	-0.54	-0.51	-0.52	+0.73	-0.014 (-0.74)	-0.35% (-1.72)		0.00	negative, not primary
A1-T tightness (Sonnet)	conf.	-0.55	-0.59	-0.58	-0.37	-0.029 (-1.42)	-0.42% (-2.06)		0.00	negative, not primary
A3 primary (Sonnet)	conf.	-0.13	-0.58	-0.44	+0.85	-0.021 (-1.08)	-0.38% (-1.86)	0.72	0.00	Do not implement
A4 islands (Sonnet)	conf.	-0.18	-0.79	-0.56	+1.23	-0.020 (-0.86)	-0.55% (-2.39)		0.00	negative, not primary
A3 (Opus follow-up)	expl.	-0.48	-0.46	-0.46	-0.64	+0.001 (+0.05)	-0.24% (-1.54)		0.00	exploratory, negative
A4 primary (Opus follow-up)	expl.	-0.40	+0.01	-0.13	-1.41	-0.002 (-0.10)	+0.01% (+0.05)	0.56	0.00	exploratory, fails G2-G5
V2a wage growth (headline)	post-hoc	+0.09	+0.16	+0.15	-1.71	+0.049 (+2.04)	+0.06% (+0.29)	0.02	0.02	noise after the looks taken
V2b signed composite	post-hoc	-0.06	-0.15	-0.11	-1.34	+0.042 (+1.65)	-0.18% (-0.84)	0.05	0.00	noise after the looks taken
V2c wage + momentum	post-hoc	+0.27	+0.35	+0.32	-0.78	+0.032 (+1.21)	+0.22% (+1.36)	0.26	0.09	noise after the looks taken
V2d -(hires+quits), 12m	post-hoc	+0.50	+0.29	+0.35	+0.29	+0.096 (+2.43)	+0.26% (+1.37)	0.85	0.06	noise after the looks taken

3 Figures

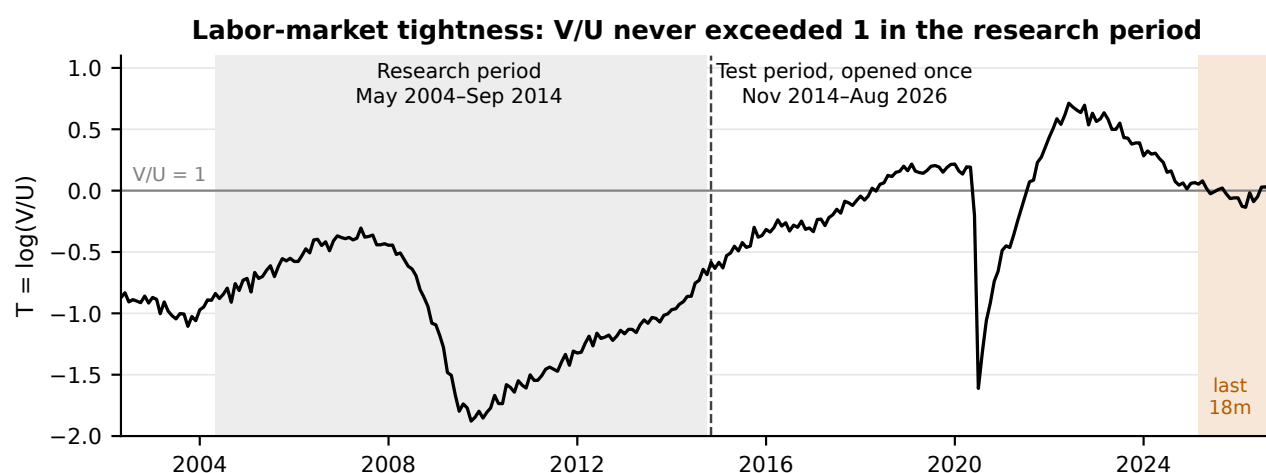


FIGURE 1. Tightness never exceeded $V/U = 1$ in the research period, so the tightness moderator could not be learned there. Source: [analysis/data/fred](#) (JTSJOL, UNEMPLOY). POST-HOC

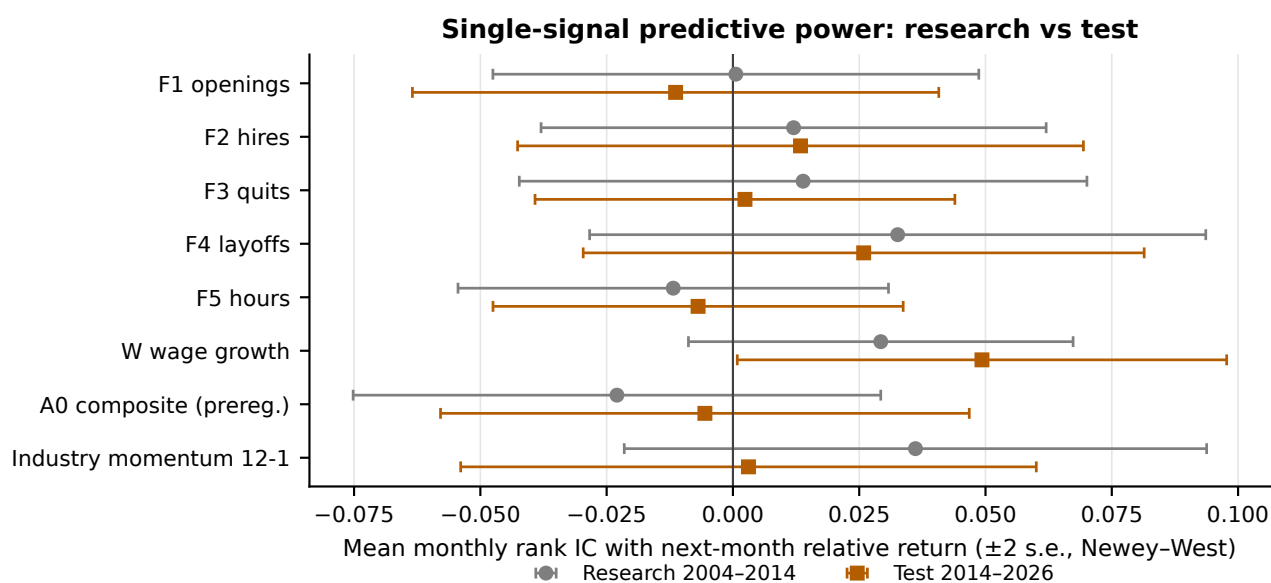


FIGURE 2. Only wage growth has a positive IC in both periods; layoffs and hours carry the opposite sign to the one A0 imposed. Source: analysis/output/tables/d1_signal_ic.csv. POST-HOC

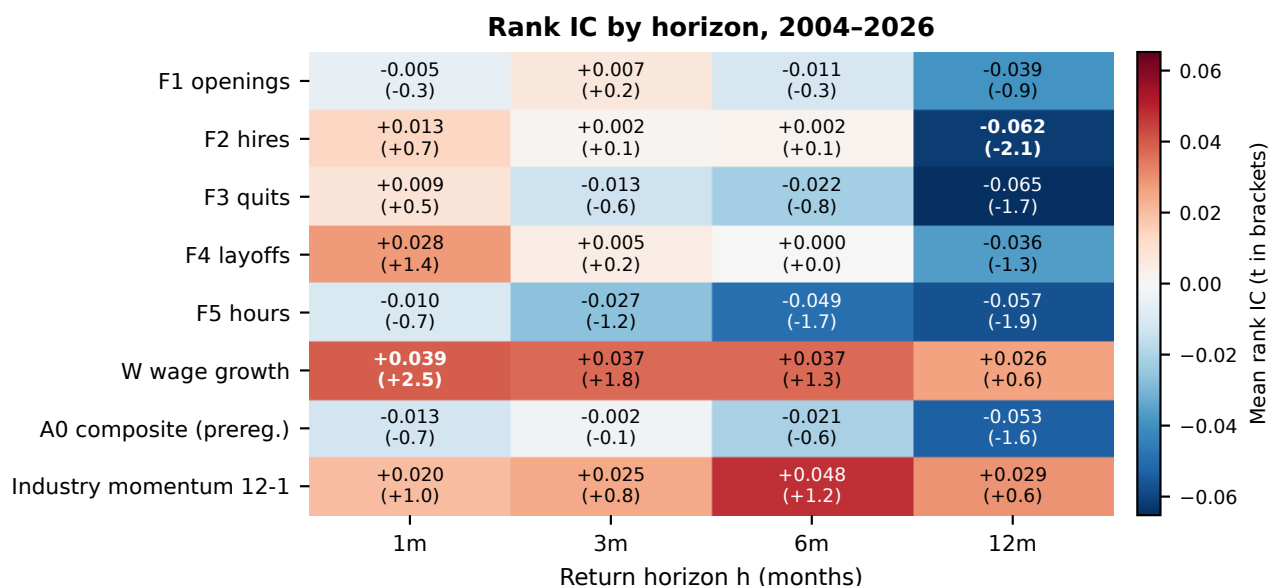


FIGURE 3. Hires and quits turn negative at 12 months (t -2.1, -1.7): the investment/cost channel shows up slowly, not at one month. Source: analysis/output/tables/d2_horizon_ic.csv. POST-HOC

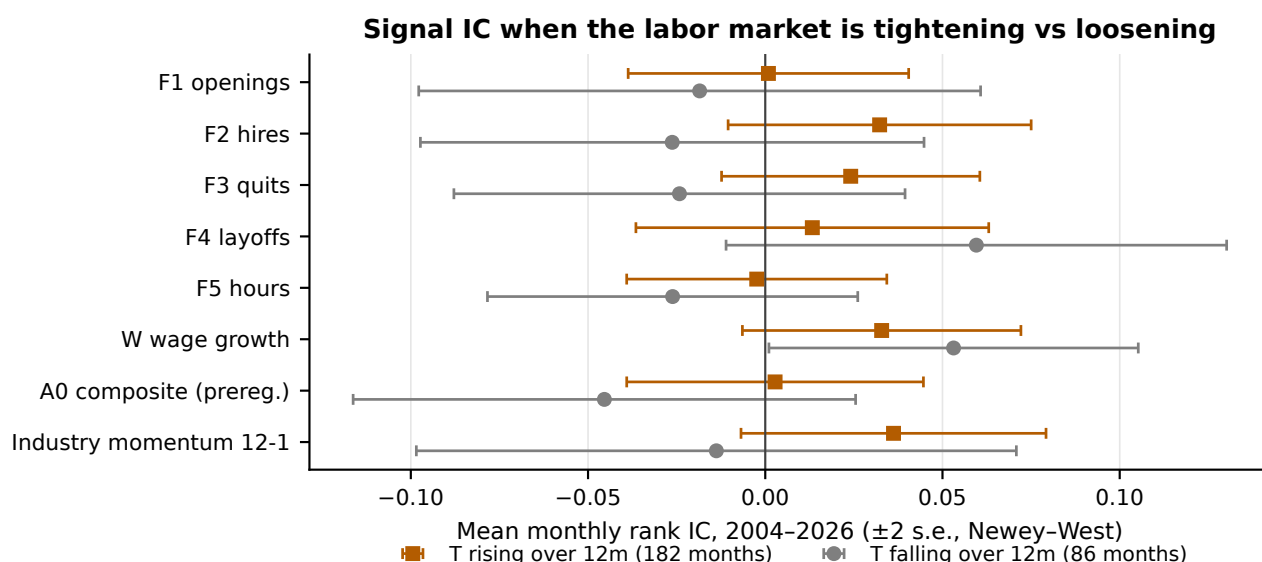


FIGURE 4. Split by whether T rose or fell over 12 months, hires and quits predict returns only while the market tightens; the split is balanced in both periods. Source: analysis/output/tables/d3_tightness_ic.csv. POST-HOC

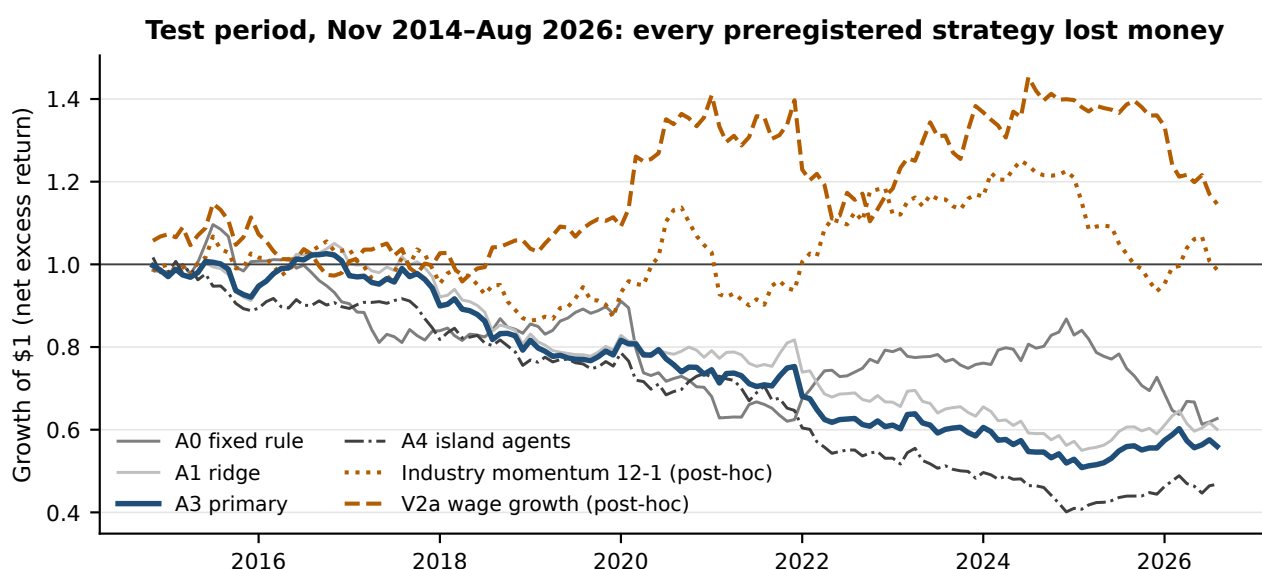


FIGURE 5. Every preregistered strategy lost money in the test; the post-hoc wage-growth book gained until 2024 and gave much of it back. Source: sealed/initial/ASOF_*_ledger.csv; analysis/output/tables/v2_monthly_net.csv. CONFIRMATORY (A0-A4) / POST-HOC (momentum, V2a)

A3 test-period P&L by indus

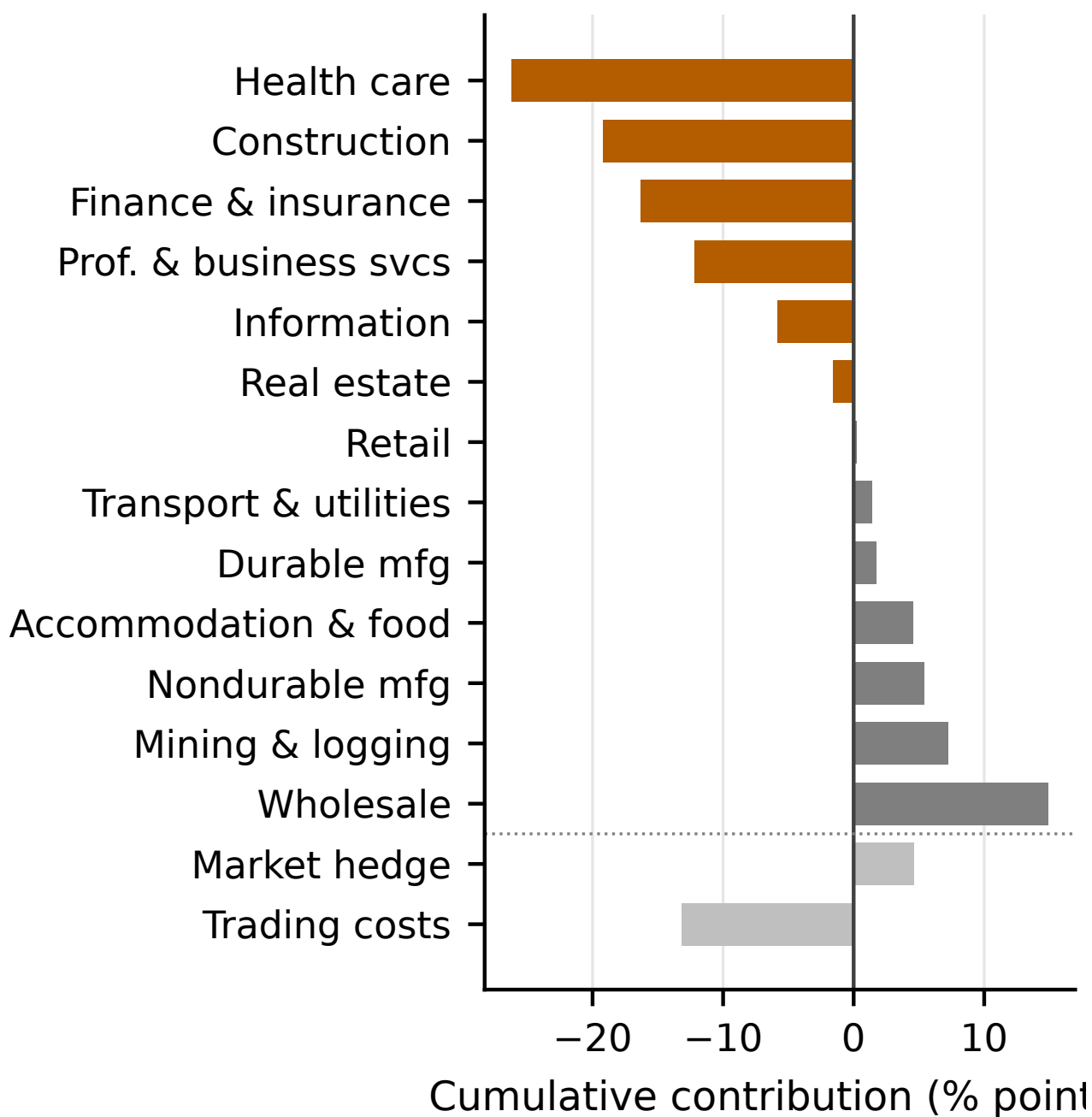


FIGURE 6. A3 lost most in health care, construction and finance; trading costs were a secondary drag. Source: analysis/output/tables/d5_industry_pnl.csv.

POST-HOC

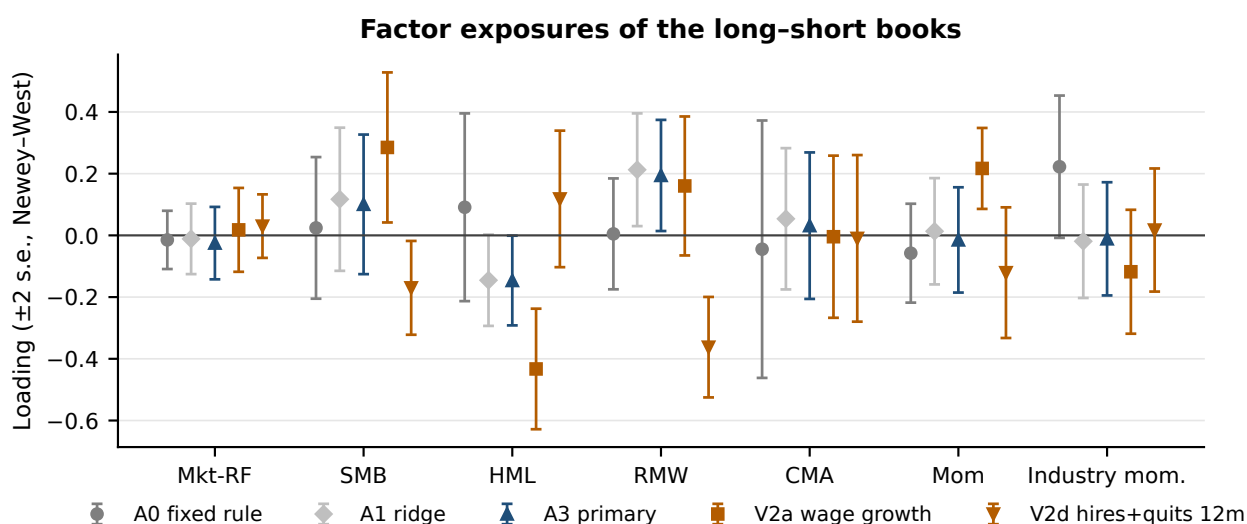


FIGURE 7. The labor books are quality/growth tilts: A3 loads on RMW and against HML; V2a loads against HML and on momentum. Source: results.json attribution; analysis/output/tables/v2a_factor_loadings_test.csv.

CONFIRMATORY (A0, A1, A3) / POST-HOC (V2a)

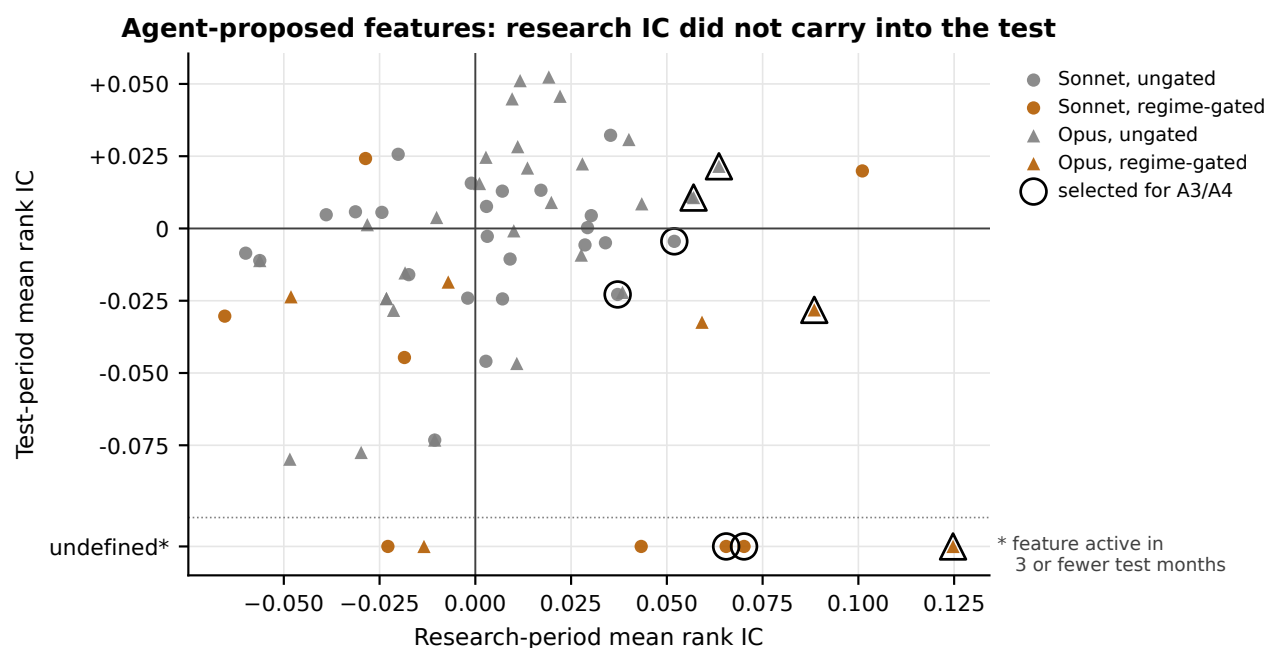


FIGURE 8. Research IC of agent features barely predicts their test IC; the selected regime-gated features were active in 3 test months. Source: analysis/output/tables/d7_agent_candidates.csv.

POST-HOC

Blinded agent experiment: two arms with equal budgets

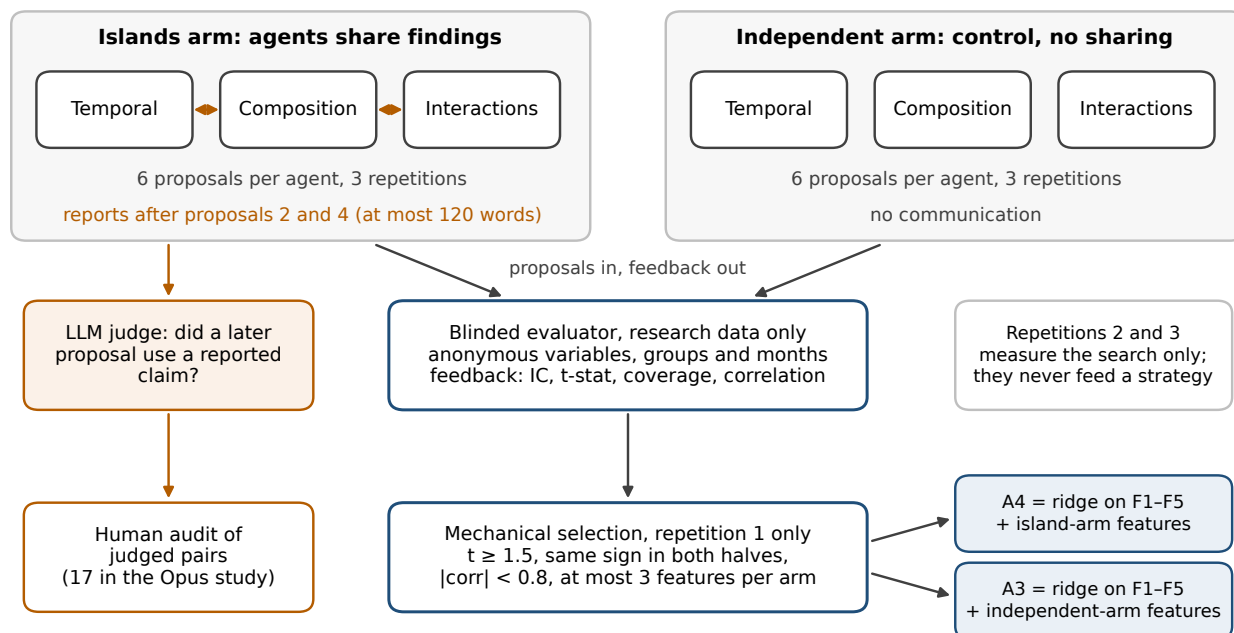


FIGURE 9. Two arms with equal budgets, a blinded evaluator, mechanical selection and a human-audited LLM judge. Source: frozen params.py AGENT. CONFIRMATORY

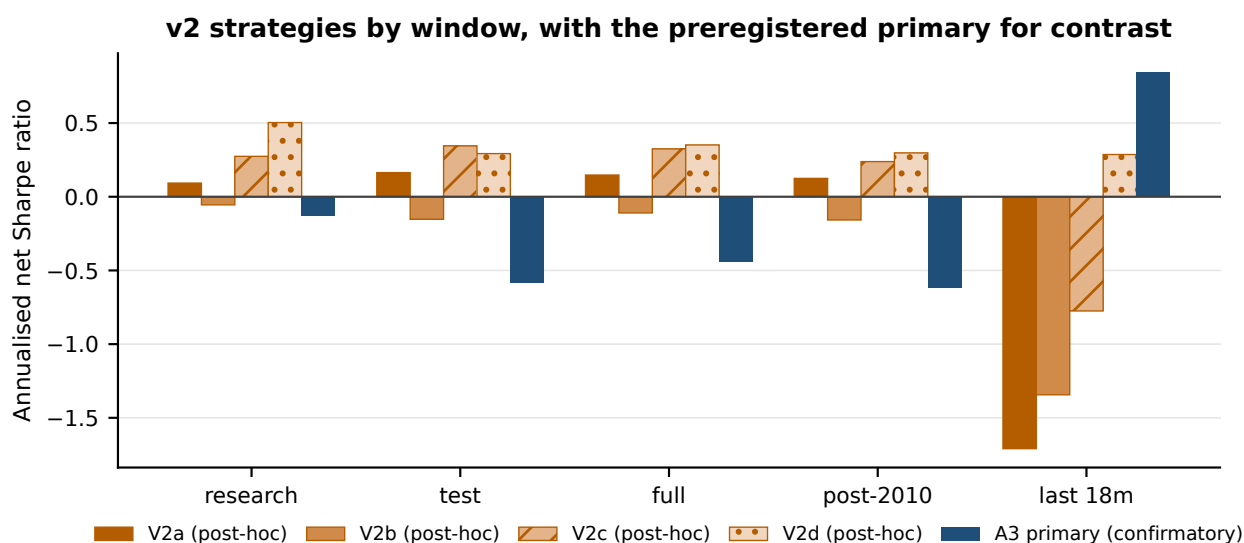


FIGURE 10. V2d is the only variant positive in every window; V2a and V2c lost heavily in the last 18 months. Source: analysis/output/tables/v2_window_stats.csv. POST-HOC / CONFIRMATORY (A3 bars)

4 Statistics

Each table lives in `analysis/output/tables/<name>.csv`; the LaTeX fragment in `report/tables/<name>.tex`.

D1 single-signal rank IC

post-hoc

Signal	Research 2004–14	Test 2014–26	Full 2004–26
F1 openings	+0.001 (+0.02)	−0.011 (−0.44)	−0.005 (−0.30)
F2 hires	+0.012 (+0.48)	+0.013 (+0.48)	+0.013 (+0.71)
F3 quits	+0.014 (+0.49)	+0.002 (+0.11)	+0.009 (+0.50)
F4 layoffs	+0.033 (+1.07)	+0.026 (+0.93)	+0.028 (+1.38)
F5 hours	−0.012 (−0.55)	−0.007 (−0.34)	−0.010 (−0.67)
W wage growth	+0.029 (+1.54)	+0.049 (+2.04)	+0.039 (+2.50)
A0 composite (prereg.)	−0.023 (−0.88)	−0.006 (−0.21)	−0.013 (−0.69)
Industry momentum 12-1	+0.036 (+1.25)	+0.003 (+0.11)	+0.020 (+0.99)

D2 IC by horizon

post-hoc

Signal	1m	3m	6m	12m
F1 openings	−0.005 (−0.31)	+0.007 (+0.23)	−0.011 (−0.28)	−0.039 (−0.92)
F2 hires	+0.013 (+0.70)	+0.002 (+0.07)	+0.002 (+0.08)	−0.062 (−2.08)
F3 quits	+0.009 (+0.50)	−0.013 (−0.55)	−0.022 (−0.78)	−0.065 (−1.73)
F4 layoffs	+0.028 (+1.37)	+0.005 (+0.17)	+0.000 (+0.01)	−0.036 (−1.28)
F5 hours	−0.010 (−0.67)	−0.027 (−1.24)	−0.049 (−1.69)	−0.057 (−1.86)
W wage growth	+0.039 (+2.51)	+0.037 (+1.77)	+0.037 (+1.28)	+0.026 (+0.63)
A0 composite (prereg.)	−0.013 (−0.69)	−0.002 (−0.09)	−0.021 (−0.61)	−0.053 (−1.58)
Industry momentum 12-1	+0.020 (+0.99)	+0.025 (+0.80)	+0.048 (+1.24)	+0.029 (+0.63)

D3a tightness regime balance

post-hoc

Months in regime	Research 2004–14	Test 2014–26
months	125	142
expanding low	49	3
expanding middle	26	6
expanding high	50	133
T rising (12m)	93	88
T falling (12m)	32	54
V/U > 1	0	75
T min	−1.88	−1.61
T max	−0.30	+0.71

D3b IC when T rises vs falls

post-hoc

Signal	T rising (182 m)	T falling (86 m)	Difference
F1 openings	+0.001 (+0.05)	−0.019 (−0.47)	+0.019
F2 hires	+0.032 (+1.51)	−0.026 (−0.74)	+0.059
F3 quits	+0.024 (+1.32)	−0.024 (−0.76)	+0.048
F4 layoffs	+0.013 (+0.53)	+0.060 (+1.69)	−0.046
F5 hours	−0.002 (−0.13)	−0.026 (−1.00)	+0.024
W wage growth	+0.033 (+1.67)	+0.053 (+2.04)	−0.020
A0 composite (prereg.)	+0.003 (+0.13)	−0.045 (−1.28)	+0.048
Industry momentum 12-1	+0.036 (+1.68)	−0.014 (−0.33)	+0.050

D4 test-period performance

confirmatory (A0-A4) / post-hoc (momentum)

strategy	Sharpe	Net p.a.	Vol.	Max DD	Turnover	Costs p.a.
A0 fixed rule	−0.36	−3.5%	9.7%	−40.3%	0.83	1.01%
A1 ridge	−0.51	−4.0%	7.9%	−37.7%	0.91	1.11%
A3 primary	−0.58	−4.6%	7.8%	−41.1%	0.91	1.11%
A4 island agents	−0.79	−6.1%	7.8%	−53.3%	1.11	1.35%
Industry momentum 12-1	+0.03	+0.3%	9.4%	−20.8%	0.53	0.65%

D5 A3 P&L by industry

post-hoc

component	Contribution (pp)	Months long	Months short
Mining & logging	+7.2	33%	36%
Construction	−19.2	36%	39%
Durable mfg	+1.8	31%	24%
Nondurable mfg	+5.4	32%	35%
Wholesale	+14.9	31%	42%
Retail	+0.2	39%	35%
Transport & utilities	+1.4	47%	30%
Information	−5.8	30%	37%
Finance & insurance	−16.3	34%	45%
Real estate	−1.6	24%	45%
Prof. & business svcs	−12.2	18%	47%
Health care	−26.2	41%	36%
Accommodation & food	+4.5	46%	22%
Market hedge	+4.6		
Trading costs	−13.1		

D6 factor loadings

confirmatory

Factor	A0 fixed rule	A1 ridge	A3 primary
Alpha (per month)	−0.25% (−0.92)	−0.35% (−1.72)	−0.38% (−1.86)
Market	−0.01 (−0.31)	−0.01 (−0.20)	−0.03 (−0.43)
SMB	+0.02 (+0.21)	+0.12 (+1.01)	+0.10 (+0.89)
HML	+0.09 (+0.60)	−0.15 (−1.97)	−0.15 (−2.01)
RMW	+0.00 (+0.05)	+0.21 (+2.33)	+0.19 (+2.16)
CMA	−0.04 (−0.21)	+0.05 (+0.47)	+0.03 (+0.26)
Momentum	−0.06 (−0.72)	+0.01 (+0.15)	−0.01 (−0.17)
Industry momentum	+0.22 (+1.93)	−0.02 (−0.21)	−0.01 (−0.12)

D7a agent candidates

post-hoc

Repetition-1 candidates	Sonnet	Opus
valid candidates	33.0	35.0
regime-gated	9.0	6.0
test IC undefined (≤ 3 active months)	4.0	2.0
corr(research IC, test IC)	0.22	0.37
passed research filter	4.0	7.0
selected	4.0	4.0
selected and regime-gated	2.0	2.0
selected with positive test IC	0.0	2.0

D7b selected agent features

post-hoc

Study	Arm	Feature (decoded)	Research IC (n)	Test IC (n)
Sonnet	independent	regime(yoy(Earnings), "low")	+0.065 (48)	n/a (3)
Sonnet	independent	regime(spread(tsz(Layoffs, 60), tsz(Hires, 60)), "low")	+0.070 (43)	n/a (3)
Sonnet	islands	spread(chg(Layoffs, 3), chg(Openings, 3))	+0.037 (123)	−0.023 (142)
Sonnet	islands	spread(chg(Layoffs, 6), chg(Openings, 6))	+0.052 (120)	−0.004 (142)
Opus	independent	mult(regime(tsz(Hires, "exp"), "high"), cstrank(tsz(Openings, "exp")))	+0.088 (15)	−0.028 (133)
Opus	islands	tsz(lchg(lag(Earnings, 1), 12), 36)	+0.064 (78)	+0.022 (142)
Opus	islands	regime(spread(chg(Quits, 3), chg(Hires, 3)), "low")	+0.125 (48)	n/a (3)
Opus	islands	tsz(lchg(lag(Hours, 1), 12), 36)	+0.057 (78)	+0.011 (142)

D8 A3 by window

confirmatory

window	Months	Sharpe	Net return p.a.	Mean IC (t)
Full test, Nov 2014-Aug 2026	142	−0.58	−4.6%	−0.021 (−1.08)
First half	71	−0.72	−4.9%	−0.028 (−0.90)
Second half	71	−0.49	−4.3%	−0.015 (−0.59)
Excluding Mar-Dec 2020	132	−0.51	−4.1%	−0.012 (−0.56)
Last 18 months (Mar 2025-Aug 2026)	18	+0.85	+6.7%	+0.033 (+0.44)
Course window "full" (from Jun 2007)	231	−0.44	−3.2%	−0.009 (−0.49)
Course window "post-2010"	200	−0.62	−4.4%	−0.019 (−1.02)

D9 risk overlay

post-hoc

Study, strategy	Cap binds	Ex-ante vol	Realised vol	Gross leverage	Hedge
Sonnet A0	80%	8.0%	9.7%	1.96	0.15
Sonnet A1	88%	7.5%	7.9%	1.98	0.13
Sonnet A1-T	82%	7.5%	8.2%	1.97	0.12
Sonnet A3	87%	7.5%	7.8%	1.98	0.13
Sonnet A4	91%	7.3%	7.8%	1.99	0.13
Opus A3	92%	7.3%	7.5%	1.99	0.15
Opus A4	93%	7.3%	8.4%	1.99	0.11

D10 agent experiment design

confirmatory

study	item	value
Both	Arms	islands (communicating), independent (control)
Both	Agents per arm	Temporal, Composition, Interactions
Both	Proposals per agent x repetitions	6 x 3
Both	Migration reports	after proposals 2 and 4, at most 120 words
Both	Selection (repetition 1 only)	$t \geq 1.5$, same sign in both halves, $ \text{corr} < 0.8$, at most 3 per arm
Sonnet	Valid proposals	98 / 108
Sonnet	Migration reports admitted	7 / 18
Sonnet	Defined LLM-judge uptake labels	8 / 36
Sonnet	Human-audited judge pairs	8
Opus	Valid proposals	106 / 108
Opus	Migration reports admitted	15 / 18
Opus	Defined LLM-judge uptake labels	84 / 88
Opus	Human-audited judge pairs	17
Opus	Human-judge agreement	10 / 17 (human yes 3, judge yes 10)

v2 variants: Sharpe by window and test statistics

post-hoc

Variant	Research	Test	Full	Post-2010	Last 18m	Test IC (t)	Alpha t	Placebo p	DSR
V2a	+0.09	+0.16	+0.15	+0.12	−1.71	+0.049 (+2.04)	+0.29	0.02	0.02
V2b	−0.06	−0.15	−0.11	−0.16	−1.34	+0.042 (+1.65)	−0.84	0.05	0.00
V2c	+0.27	+0.35	+0.32	+0.24	−0.78	+0.032 (+1.21)	+1.36	0.26	0.09
V2d	+0.50	+0.29	+0.35	+0.30	+0.29	−0.020 (−0.87)	+1.37	0.85	0.06

5 V2d robustness battery (v3)

Spec analysis/v3_spec.md frozen 2026-10-03T09:52:27Z, first run 09:53:32Z; every check run once; N_{trials} raised to 132. Results: analysis/output/v3_results.md. POST-HOC

Reading against the pre-declared rule: fragile in one dimension (R_4). V2d lost money in the first half of the test (Sharpe -0.21) and made it in the second ($+0.64$); the other six conditions hold. The 12-month IC is positive in both halves.

R1 Components

post-hoc

Sharpe research	Sharpe test	Sharpe full	Test IC 12m (t)	Test alpha t
+0.41	+0.19	+0.27	+0.093 (+2.90)	+0.79
+0.18	+0.16	+0.15	+0.085 (+1.75)	+0.81
+0.10	+0.42	+0.29	+0.082 (+1.38)	+1.48
+0.11	+0.16	+0.13	+0.032 (+0.87)	+0.69
+0.50	+0.29	+0.35	+0.096 (+2.43)	+1.37

R2 Horizon profile

post-hoc

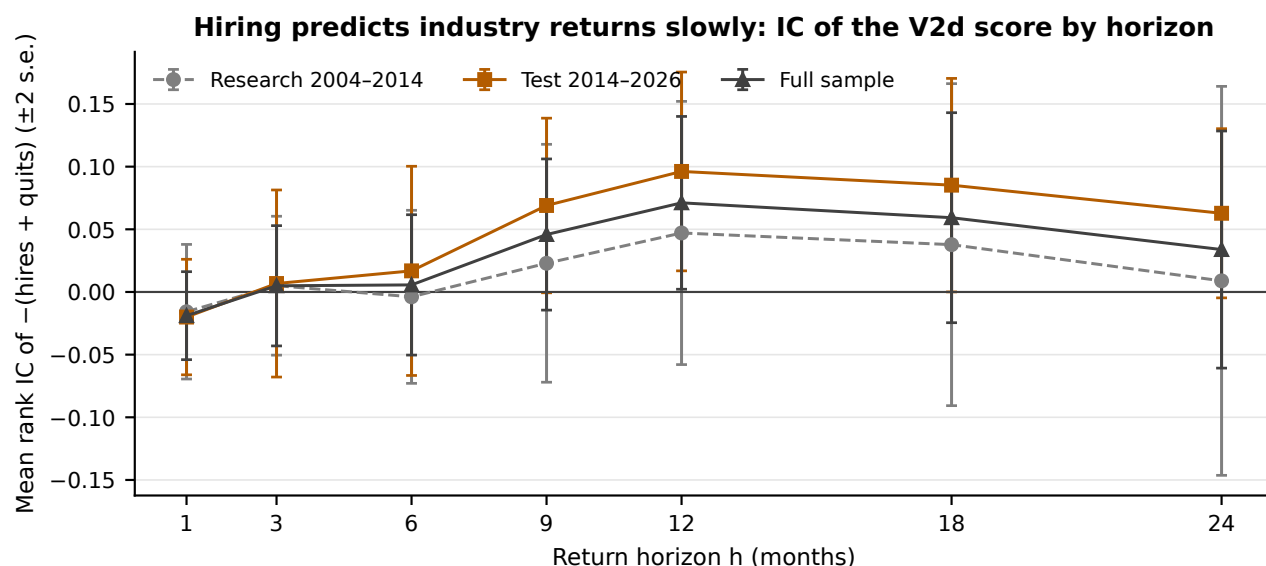


FIGURE 11. The V2d score has no one-month IC and a positive IC from 9 to 24 months, peaking at 12 (test $+0.096$, $t = 2.43$). Source: analysis/output/tables/v3_r2_horizon_profile.csv. POST-HOC

research	test	full
-0.016 (-0.59)	-0.020 (-0.87)	-0.019 (-1.08)
+0.005 (+0.18)	+0.007 (+0.18)	+0.005 (+0.21)
-0.004 (-0.11)	+0.017 (+0.40)	+0.006 (+0.20)
+0.023 (+0.48)	+0.069 (+1.98)	+0.046 (+1.52)
+0.047 (+0.90)	+0.096 (+2.43)	+0.071 (+2.06)
+0.038 (+0.59)	+0.085 (+2.00)	+0.059 (+1.41)
+0.009 (+0.11)	+0.063 (+1.86)	+0.034 (+0.72)

R3 Holding period

post-hoc

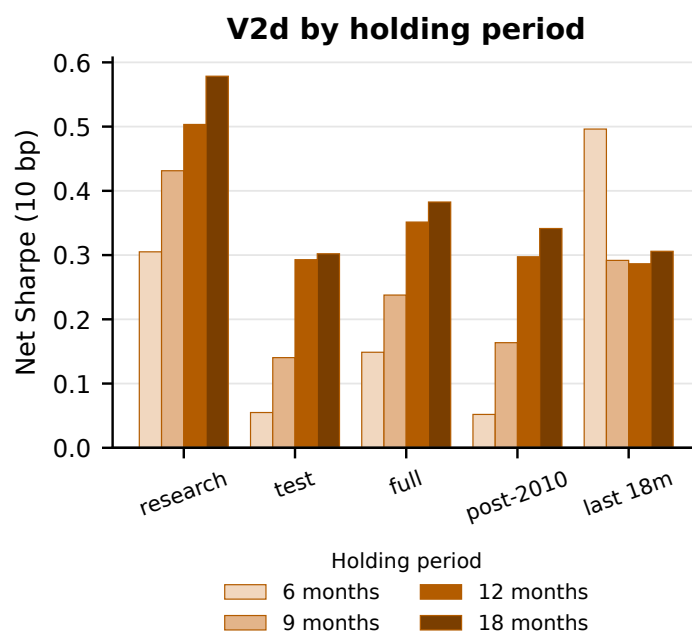


FIGURE 12. Longer holding helps: test Sharpe +0.05 at 6 months, +0.29 at 12, +0.30 at 18. Source: v3_r3_holding.csv.

POST-HOC

Holding	research	test	full	post-2010	last 18m	Test turnover
6 months	+0.31	+0.05	+0.15	+0.05	+0.50	0.49
9 months	+0.43	+0.14	+0.24	+0.16	+0.29	0.40
12 months	+0.50	+0.29	+0.35	+0.30	+0.29	0.37
18 months	+0.58	+0.30	+0.38	+0.34	+0.31	0.28

R4 Sub-periods

post-hoc

Window	Months	Sharpe	Net p.a.	IC 12m (t)
test: first half	71	-0.21	-1.4%	+0.097 (+1.75)
test: second half	71	+0.64	+6.5%	+0.095 (+1.60)
test: excl. Mar 2020-Dec 2021	120	+0.37	+3.0%	+0.105 (+2.23)
test: pre-2020	62	+0.14	+0.8%	+0.114 (+1.92)
test: 2020 onward	80	+0.37	+3.9%	+0.080 (+1.54)
full: first half	134	+0.33	+2.0%	+0.038 (+0.73)
full: second half	134	+0.38	+3.3%	+0.107 (+2.66)
full: excl. Mar 2020-Dec 2021	246	+0.40	+2.9%	+0.073 (+1.93)
full: pre-2020	188	+0.36	+2.2%	+0.068 (+1.57)

R5 Costs

post-hoc

Window	0 bp	10 bp	25 bp	50 bp	Break-even (bp)
research	+0.58	+0.50	+0.39	+0.20	76
test	+0.34	+0.29	+0.22	+0.09	68
full	+0.41	+0.35	+0.26	+0.11	69
post-2010	+0.35	+0.30	+0.21	+0.07	63
last 18m	+0.32	+0.29	+0.24	+0.16	99

R6 Factor regression, 12 HAC lags

post-hoc

Test (HAC 12)	Full (HAC 12)
+0.26% (+1.48)	+0.26% (+1.92)
+0.03 (+0.58)	−0.00 (−0.08)
−0.17 (−2.24)	−0.02 (−0.31)
+0.12 (+1.07)	+0.10 (+1.23)
−0.36 (−4.45)	−0.16 (−1.83)
−0.01 (−0.07)	−0.09 (−0.87)
−0.12 (−1.14)	−0.03 (−0.51)
+0.02 (+0.17)	+0.06 (+0.93)

R7 Drop one industry

post-hoc

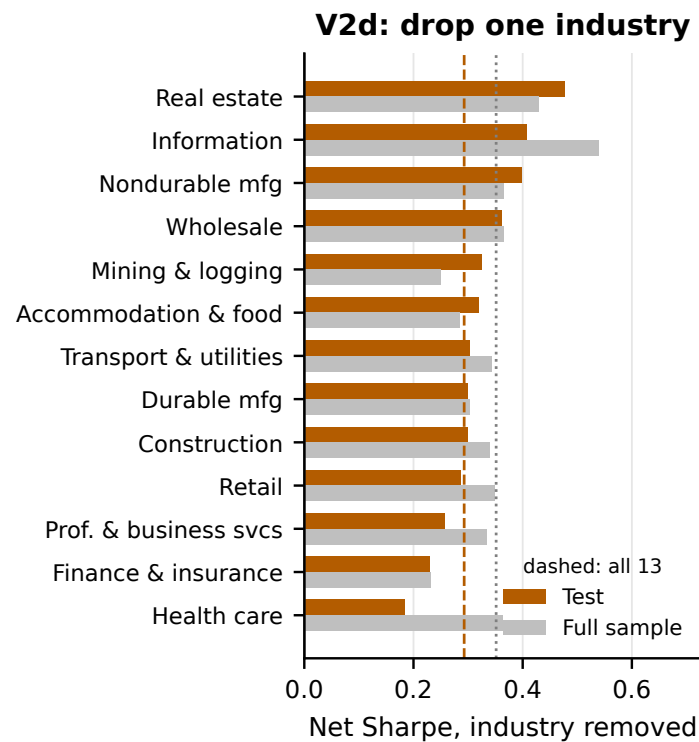


FIGURE 13. No single industry carries V2d: removing any one leaves the test Sharpe between +0.18 and +0.48. Source: v3_r7_dropone.csv.

POST-HOC

statistic	Test Sharpe	Full Sharpe
minimum	+0.18	+0.23
median	+0.30	+0.34
maximum	+0.48	+0.54
all 13 groups (V2d)	+0.29	+0.35
industry whose removal lowers test Sharpe most	Health care	
industry whose removal raises test Sharpe most	Real estate	

R8 Tightness moderator

post-hoc

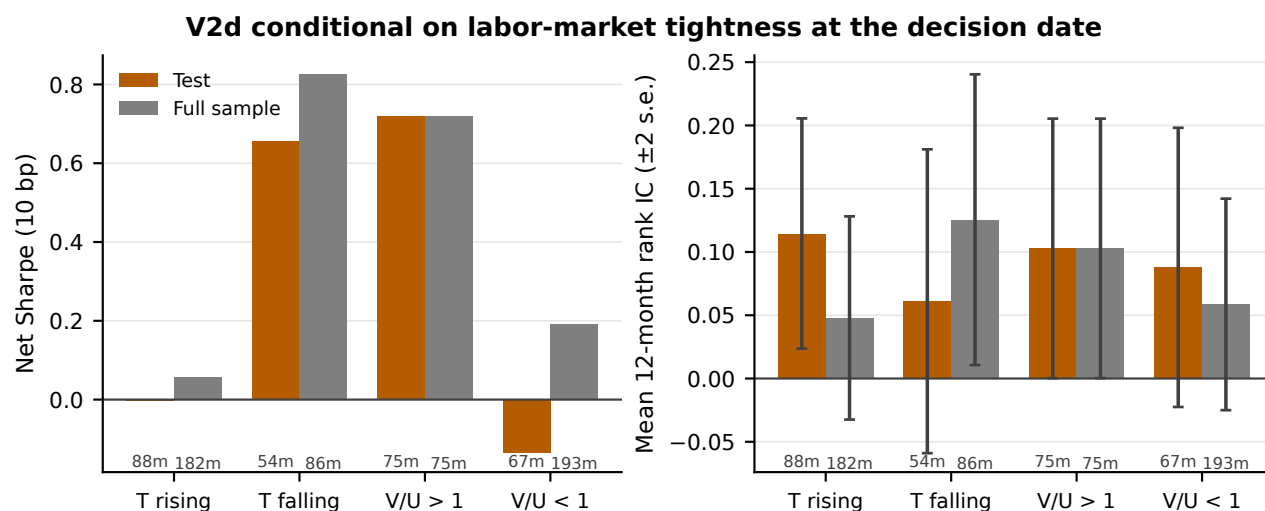


FIGURE 14. V2d earned its return when tightness was falling or $V/U > 1$; in tightening months the Sharpe was zero although the 12-month IC was highest. Source: v3_r8_tightness.csv.

POST-HOC

Months	Sharpe	Net p.a.	IC 12m (t)
88	-0.00	-0.0%	+0.115 (+2.52)
54	+0.66	+6.8%	+0.061 (+1.02)
75	+0.72	+6.0%	+0.103 (+2.00)
67	-0.14	-1.2%	+0.088 (+1.59)
182	+0.06	+0.4%	+0.048 (+1.19)
86	+0.83	+7.6%	+0.126 (+2.19)
75	+0.72	+6.0%	+0.103 (+2.00)
193	+0.19	+1.4%	+0.059 (+1.40)

R9 Block-bootstrap interval

post-hoc

Window	Sharpe	90% interval	Months	Excludes 0
test	+0.29	[-0.07, +0.65]	142	no
full	+0.35	[+0.05, +0.64]	268	yes

R10 As-known data

post-hoc

check	status	note
R10 as-known (ALFRED vin- tage) rerun of V2a-V2d	not run: no key	analysis/.env has no FRED_API_KEY; historical v2/v3 re- sults use revised data with fixed lags (data_vintage.md)

R11 Fundamental law, test window

confirmatory (A3) / post-hoc

Book (test window)	IC	Horizon	BR/yr	TC	Implied IR	Ceiling (TC=1)	Realised Sharpe
A3 primary	-0.021	1m	156	0.80	-0.21	-0.27	-0.58
V2a wage growth	+0.049	1m	156	0.91	+0.56	+0.62	+0.16
V2d -(hires+quits) 12m	+0.096	12m	13	0.48	+0.17	+0.35	+0.29

Deflated Sharpe at 112 and 132 trials**post-hoc**

Window	DSR (112 trials)	DSR (132 trials)	Benchmark Sharpe (132)
test	0.02	0.02	0.76
full	0.03	0.03	0.56
test	0.00	0.00	0.76
full	0.00	0.00	0.56
test	0.09	0.08	0.76
full	0.16	0.15	0.56
test	0.06	0.05	0.76
full	0.18	0.16	0.56

Reading rule**post-hoc**

condition	holds
R1 hires-only and quits-only test Sharpe > 0	yes
R2 test 12m IC $t \geq 2$ and 1m IC $t < 2$	yes
R3 holding 9 and 18 months test Sharpe > 0	yes
R4 both test halves Sharpe > 0	no
R5 test break-even cost > 25 bp	yes
R7 minimum drop-one test Sharpe > 0	yes
R9 90% interval excludes 0 (test or full)	yes

6 The agent experiment

study	item	value
Both	Arms	islands (communicating), independent (control)
Both	Agents per arm	Temporal, Composition, Interactions
Both	Proposals per agent x repetitions	6 x 3
Both	Migration reports	after proposals 2 and 4, at most 120 words
Both	Selection (repetition 1 only)	$t \geq 1.5$, same sign in both halves, $ \text{corr} < 0.8$, at most 3 per arm
Sonnet	Valid proposals	98 / 108
Sonnet	Migration reports admitted	7 / 18
Sonnet	Defined LLM-judge uptake labels	8 / 36
Sonnet	Human-audited judge pairs	8
Opus	Valid proposals	106 / 108
Opus	Migration reports admitted	15 / 18
Opus	Defined LLM-judge uptake labels	84 / 88
Opus	Human-audited judge pairs	17
Opus	Human-judge agreement	10 / 17 (human yes 3, judge yes 10)

Selected features and their research→test fate:

Study	Arm	Feature (decoded)	Research IC (n)	Test IC (n)
Sonnet	independent	regime(yoy(Earnings), "low")	+0.065 (48)	n/a (3)
Sonnet	independent	regime(spread(tsz(Layoffs, 60), tsz(Hires, 60)), "low")	+0.070 (43)	n/a (3)
Sonnet	islands	spread(chg(Layoffs, 3), chg(Openings, 3))	+0.037 (123)	−0.023 (142)
Sonnet	islands	spread(chg(Layoffs, 6), chg(Openings, 6))	+0.052 (120)	−0.004 (142)
Opus	independent	mult(regime(tsz(Hires, "exp"), "high"), csrank(tsz(Openings, "exp")))	+0.088 (15)	−0.028 (133)
Opus	islands	tsz(lchg(lag(Earnings, 1), 12), 36)	+0.064 (78)	+0.022 (142)
Opus	islands	regime(spread(chg(Quits, 3), chg(Hires, 3)), "low")	+0.125 (48)	n/a (3)
Opus	islands	tsz(lchg(lag(Hours, 1), 12), 36)	+0.057 (78)	+0.011 (142)

Human vs judge (Opus): 10/17 agreement; human yes 3, judge yes 10; all 7 disagreements judge-yes/human-no. Models from the logs: Sonnet 5.5 at medium effort for the confirmatory study (607 calls), Opus 5.5 at xhigh for the follow-up (905 calls).

7 AI-interaction inventory

ID	Purpose	Model	Prompt file	What was right	What was wrong	What changed	Team critique
P1	Source and code audit before research	claude-sonnet-5-5 (medium); Opus study: claude-opus-5-5 (xhigh)	outputs/follow_the_workers/outputs/p1_request.json	25 numbered findings; found the shared CES series for groups 9/10 and several metadata inconsistencies.	Several alleged timing errors were not supported (release filtering preceeds the month bound); no numeric change resulted.	Mapping meta-data corrected (prescribed vs verified scope); construction SA exception flagged; no basket or feature change.	
P2	Blinded signal-search agents (two arms, 3 repetitions)	claude-sonnet-5-5 (medium); Opus study: claude-opus-5-5 (xhigh)	params.py SYSTEM_PROMPT / TURN_PROMPT; outputs/agents/log.jsonl	Rediscovered wage growth (V05) blind in both studies; produced valid grammar expressions (98/108 Sonnet, 106/108 Opus).	Highest research t-statistics came from regime gates that were active in 3 test months; every selected feature failed the test.	Mechanical selection only; no engineer edits. v2 restores wage growth as a public, ungated signal.	
P3	Migration reports and LLM judge of idea uptake	claude-sonnet-5-5 (medium); Opus study: claude-opus-5-5 (xhigh)	outputs/agents/migrations.json; judge prompt in agents.py trace_migrations	Opus: 15/18 reports admitted and 84/88 usable labels; a human audit of 17 pairs was possible.	Sonnet: 28/36 judge outputs were not bare JSON; judge over-attributes uptake (10 yes vs human 3; all 7 disagreements judge-yes).	Judge output treated as an association, not evidence; human labels reported alongside.	
P4	Robustness red team on research tables	claude-sonnet-5-5 (medium); Opus study: claude-opus-5-5 (xhigh)	outputs/follow_the_workers/outputs/p4_request.json	Ten ranked concerns; called "do not implement" from research tables before the seal opened.	Asserted uncomputed p-values and proposed unregistered thresholds; every suggestion was already covered by a registered diagnostic.	No new test adopted (all dispositions "covered by" or "not adopted"); registered diagnostics retained.	
P5	Post-mortem review of the frozen study	claude-opus-5-5 (Claude Code, 3 Oct 2026)	conversation request (Piero, 3 Oct 2026); FINDINGS_AND_PROPOSAL.md	Found the wrong A0 signs, the dropped wage signal, the 12-month hiring sign, the 3-of-142-month agent features, the one-bin tightness split.	Guessed the engineering assistant was OpenAI Codex (unverified); used a momentum definition that included month m-1; quoted W ICs with a non-frozen start date.	Momentum redefined on m-12..m-2; W ICs recomputed with the frozen feature start; Codex guess withdrawn (to be confirmed by Alex).	
P6	This iteration: rebuild, diagnostics, frozen v2/v3 specs, forward test, (overnight) report	claude-opus-5-5 (Parts 1-3); claude-fable-5-1 (overnight)	ITERATION_PROMPT.md, OVERNIGHT_PROMPT.md, CLAUDE.md	Reproduced the sealed A0 returns to 1e-16; froze specs before running; reported all variants; stopped at a failed gate instead of working around it.	The forward spec assumed August JOLTS would not be out by 29 Sep; it was released that day, so the pre-committed check stopped the run.	Amendment 001 (fixed-lag rule, V2c/V2d added, disclosed); run log guarantees v2 outputs never changed.	

Output files: P1: p1_response.md, p1_review.md; P2: candidates.csv, agents/candidate_records.json; P3: agents/migration_judgments.json, human_review_submission.md; P4: p4_response.md, p4_

review.md; P5: FINDINGS_AND_PROPOSAL.md, analysis/output/diagnostics.md; P6: analysis/run.ipynb, analysis/output/*.md, results/, report/main.tex.

8 Claims register

88 claims, 88 verified. Source: `results/claims.csv`.

ID	Label	Claim	Numbers	Ver.
C01	CONF	The preregistered primary A3 earned a test Sharpe of -0.58 over 142 months (Nov 2014-Aug 2026).	-0.58; 142	y
C02	CONF	A3 lost 4.6% a year net of costs, with 7.8% volatility and a 41% maximum drawdown.	-4.6; 7.8; -41	y
C03	CONF	A3 one-month rank IC in the test was -0.021 (Newey-West $t = -1.08$).	-0.021; -1.08	y
C04	CONF	A3 factor alpha was -0.38% a month after FF5, momentum and industry momentum.	-0.38	y
C04b	CONF	The t-statistic of the A3 alpha was -1.86.	-1.86	y
C05	CONF	The circular-shift placebo p-value for A3 was 0.72.	0.72	y
C06	CONF	A3 break-even trading cost was -32 bp: it loses money even with free industry trading.	-32	y
C07	CONF	A3 passed gate G1 and failed G2, G3, G4 and G5.	True; False; False; False; False	y
C08	CONF	Test Sharpe ratios: A0 -0.36, A1 -0.51, A1-T -0.59, A3 -0.58, A4 -0.79.	-0.36; -0.51; -0.59; -0.58; -0.79	y
C09	CONF	Research-period Sharpe ratios (88 common months): A0 -0.44, A1 -0.54, A1-T -0.55, A3 -0.13, A4 -0.18.	88; -0.44; -0.54; -0.55; -0.13; -0.18	y
C10	CONF	A3 by window: first half -0.72, second half -0.49, excluding Mar-Dec 2020 -0.51, last 18 months +0.85.	-0.72; -0.49; -0.51; 0.85	y
C11	CONF	A3 descriptive course windows: full sample (from Jun 2007) -0.44, post-2010 -0.62.	-0.44; -0.62	y
C12	CONF	A3 Sharpe at 25 bp industry costs was -0.79 and at zero industry costs -0.45.	-0.79; -0.45	y
C13	CONF	A3 loads on profitability (RMW +0.19, $t = 2.2$) and against value (HML -0.15, $t = -2.0$).	0.19; 2.2; -0.15; -2.0	y
C14	CONF	A0 loads +0.22 on industry momentum ($t = 1.9$).	0.22; 1.9	y
C15	CONF	A3 ran at 1.1% a year of trading costs against a gross loss of 3.5% a year.	1.11; -3.5	y
C16	CONF	All four Holm-adjusted comparisons in the Sonnet study had adjusted $p = 1.0$.	1.0; 1.0; 1.0; 1.0	y
C17	CONF	Of the 142 test months, A3 was in the expanding-tercile "high" tightness bin in 133.	133	y
C18	CONF	The cross-validated ridge penalty was 1000, the top of the grid, for every learned model.	1000; 1000; 1000; 1000	y
C19	CONF	Four of four agent features that passed research selection failed the test (Sonnet); six of six in the Opus study.	4; 6	y
X01	CONF	Opus follow-up: primary A4 test Sharpe +0.01, research Sharpe -0.40, placebo $p = 0.56$, Sharpe at 25 bp -0.19.	0.01; -0.4; 0.56; -0.19	y
X02	CONF	Opus A4 one-month IC -0.002 ($t = -0.10$); alpha $t = 0.05$; last 18 months Sharpe -1.41.	-0.002; -0.1; 0.05; -1.41	y
X03	CONF	Opus A3 test Sharpe -0.46.	-0.46	y
X04	POST-HOC	In the Opus A4 book, durable manufacturing alone contributed +0.27 of cumulative gross return in the test.	0.273	y
D01	POST-HOC	Wage growth W: rank IC +0.029 in research ($t = 1.54$), +0.049 in test ($t = 2.04$), +0.039 full sample ($t = 2.50$).	0.029; 1.54; 0.049; 2.04; 0.039; 2.5	y
D02	POST-HOC	Full-sample one-month ICs: openings -0.005, hires +0.013, quits +0.009, layoffs +0.028, hours -0.010, A0 composite -0.013, industry momentum +0.020.	-0.005; 0.013; 0.009; 0.028; -0.01; -0.013; 0.02	y
D03	POST-HOC	Layoffs (F4) carry a positive IC (+0.033 research, +0.026 test) although A0 gave them a negative sign; hours (F5) carry a negative IC (-0.012, -0.007) although A0 gave them a positive sign.	0.033; 0.026; -0.012; -0.007	y
D04	POST-HOC	At a 12-month horizon hires have IC -0.062 ($t = -2.1$) and quits -0.065 ($t = -1.7$); at one month +0.013 and +0.009.	-0.062; -2.1; -0.065; -1.7; 0.013; 0.009	y
D05	POST-HOC	Wage growth IC by horizon: +0.039 at 1 month ($t = 2.5$), +0.026 at 12 months.	0.039; 2.5; 0.026	y
D06	POST-HOC	Expanding tightness terciles put 133 of 142 test months in the "high" bin (3 low, 6 middle); V/U exceeded 1 in 0 research months and 75 test months.	133; 3; 6; 0; 75	y
D07	POST-HOC	A 12-month-change split is balanced: T rising in 93 research and 88 test months, falling in 32 and 54.	93; 88; 32; 54	y
D08	POST-HOC	Industry momentum 12-1 through the same backtest had a test Sharpe of +0.03.	0.03	y
D09	POST-HOC	A3 test P&L by industry: health care -26.2 pp, construction -19.2, finance -16.3, professional services -12.2; wholesale +14.9; trading costs -13.1.	-26.2; -19.2; -16.3; -12.2; 14.9; -13.1	y

ID	Label	Claim	Numbers	Ver.
D10	POST-HOC	Selected agent features: 2 of 4 regime-gated in each study; test IC undefined for 4 Sonnet and 2 Opus candidates; 0 (Sonnet) and 2 (Opus) selected features had a positive test IC.	2; 2; 4; 2; 0; 2	y
D11	POST-HOC	Correlation between research and test IC across valid agent candidates: 0.22 (Sonnet), 0.37 (Opus).	0.22; 0.37	y
D12	POST-HOC	Sonnet's two independent-arm features were active in only 3 of 142 test months; the best Opus island feature (research t 3.56) also in 3.	3; 3; 3.56; 3	y
D13	POST-HOC	The gross-exposure cap bound in 87% of A3 test months; ex-ante volatility averaged 7.5% against a 10% target (realised 7.8%).	87; 7.5; 7.8	y
D14	POST-HOC	Across the preregistered books the cap bound in 80% to 93% of test months.	80; 93	y
A01	CONF	Sonnet study: 98 of 108 proposals valid, 7 of 18 migration reports admitted, 8 of 36 judge labels usable, 8 pairs human-audited.	98 / 108; 7 / 18; 8 / 36; 8	y
A02	CONF	Opus study: 106 of 108 proposals valid, 15 of 18 reports admitted, 84 of 88 judge labels usable, 17 pairs human-audited, human-judge agreement 10 of 17 (human yes 3, judge yes 10).	106 / 108; 15 / 18; 84 / 88; 17; 10 / 17 (human yes 3, judge yes 10)	y
A03	CONF	All 7 human-judge disagreements were judge-yes / human-no (items 2, 3, 8, 12, 13, 16, 17).	7; 0	y
A04	CONF	Judge-labelled uptake of migrated ideas: 25% of usable Sonnet labels (2 of 8), 62% of Opus labels (52 of 84).	0.25; 0.619	y
A05	CONF	Best-so-far research IC, islands vs independent (mean over 3 runs): Sonnet 0.049 vs 0.083, Opus 0.136 vs 0.059; neither gap exceeds the seed-to-seed range.	0.049; 0.083; 0.136; 0.059; False; False	y
V__ V2a__ SR	POST-HOC	V2a net Sharpe (10 bp): research +0.09, test +0.16, full +0.15, post-2010 +0.12, last 18 months -1.71.	0.09; 0.16; 0.15; 0.12; -1.71	y
V__ V2b__ SR	POST-HOC	V2b net Sharpe (10 bp): research -0.06, test -0.15, full -0.11, post-2010 -0.16, last 18 months -1.34.	-0.06; -0.15; -0.11; -0.16; -1.34	y
V__ V2c__ SR	POST-HOC	V2c net Sharpe (10 bp): research +0.27, test +0.35, full +0.32, post-2010 +0.24, last 18 months -0.78.	0.27; 0.35; 0.32; 0.24; -0.78	y
V__ V2d__ SR	POST-HOC	V2d net Sharpe (10 bp): research +0.50, test +0.29, full +0.35, post-2010 +0.30, last 18 months +0.29.	0.5; 0.29; 0.35; 0.3; 0.29	y
V01	POST-HOC	V2a test: one-month IC +0.049 (t 2.04), alpha t 0.29, placebo p 0.02, deflated Sharpe 0.02, Sharpe at 25 bp +0.10, turnover 0.39 per month.	0.049; 2.04; 0.29; 0.02; 0.02; 0.1; 0.39	y
V02	POST-HOC	V2d test: 12-month IC +0.096 (t 2.43), one-month IC -0.020 (t -0.87), alpha t 1.37, pre-declared placebo p 0.85, deflated Sharpe 0.06, Sharpe at 25 bp +0.22.	0.096; 2.43; -0.02; -0.87; 1.37; 0.85; 0.06; 0.22	y
V03	POST-HOC	V2d full sample: alpha t 1.96, 12-month IC +0.071 (t 2.06), deflated Sharpe 0.18, pre-declared placebo p 0.86.	1.96; 0.071; 2.06; 0.18; 0.86	y
V04	POST-HOC	V2d test: net return +2.6% a year, volatility 8.7%, maximum drawdown -14.4%, hit rate 55%, turnover 0.37 per month.	2.6; 8.7; -14.4; 55; 37	y
V05	POST-HOC	V2c test: alpha t 1.36, placebo p 0.26, deflated Sharpe 0.09.	1.36; 0.26; 0.09	y
V06	POST-HOC	V2b test: one-month IC +0.042 (t 1.65) but alpha t -0.84.	0.042; 1.65; -0.84	y
V07	POST-HOC	No v2 variant met the pre-declared reading rule.	not distinguishable from noise after the looks already taken; not distinguishable from noise after the looks already taken; not distinguishable from noise after the looks already taken; not distinguishable from noise after the looks already taken	y
V08	POST-HOC	The highest deflated Sharpe among all v2 variants and windows was 0.18 (V2d, full sample).	0.18	y
V09	POST-HOC	Horizon-matched (12-month IC) placebo for V2d, added post-hoc: p 0.06 test, 0.04 full sample, 0.00 post-2010, 0.26 research.	0.06; 0.04; 0.0; 0.26	y
V10	POST-HOC	V2b signs learned on research data: + for openings, hires, quits, layoffs and wages; - for hours.	1; 1; 1; 1; 1; -1	y
V11	POST-HOC	V2a test loadings: HML -0.43 (t -4.4), momentum +0.22 (t 3.3), SMB +0.28 (t 2.3).	-0.43; -4.4; 0.22; 3.3; 0.28; 2.3	y
F01	FORWARD	V2a Q4 2026 book: long construction, durable manufacturing and information; short mining, health care and accommodation & food.	0.3333333333333333; 0.3333333333333333; 0.3333333333333333; -0.3333333333333333; -0.3333333333333333	y - -

ID	Label	Claim	Numbers	Ver.
F02	FORWARD TEST	V2d Q4 2026 book: long finance, real estate and accommodation & food; short mining, durable manufacturing and retail.	0.3333333333333333; 0.3333333333333333; 0.3333333333333333; 0.3333333333333333; -0.3333333333333333; -0.3333333333333333	y -
F03	FORWARD TEST	SPY hedge per unit tranche: V2a -0.40, V2b -0.06, V2c -0.24, V2d -0.02.	-0.4; -0.06; -0.24; -0.02	y
F04	FORWARD TEST	V2c's long durable manufacturing and short professional services both map to XLI and cancel, so its tradable book has four legs.	0.0	y
F05	FORWARD TEST	The forward spec was frozen at 2026-10-03T09:03:56Z and amendment 001 at 2026-10-03T09:16:11Z, before the book was built at 2026-10-03T09:17:55Z.	True; True	y
S01	POST-HOC	Our rebuilt A0 signal matches the frozen fixed-lag scores with pooled correlation 0.998.	0.998	y
S02	POST-HOC	As-known (ASOF) scores pick the same top three groups as our revised-data signal in only 42% of test months.	42	y
S03	POST-HOC	Durable manufacturing is 60% semiconductors (Chips) by market cap in August 2026; nondurable manufacturing is 60% pharmaceuticals.	Chips 60%, Mach 9%, Autos 7%; Drugs 60%, Hshld 9%, Soda 7%	y
S04	CONF	Gate 1: the 13 group returns match all 50 saved ledgers to 1.5e-16 and our estimators match results.json to 3.6e-15.	1.51e-16; 3.55e-15	y
R1a	POST-HOC	Held 12 months, hires alone earn a test Sharpe of +0.19 and quits alone +0.16; openings +0.42 and layoffs +0.16 (both with the same negative sign, for contrast).	0.19; 0.16; 0.42; 0.16	y
R1b	POST-HOC	Hires alone have a test 12-month IC of +0.093 (t 2.90); quits alone +0.085 (t 1.75).	0.093; 2.9; 0.085; 1.75	y
R2a	POST-HOC	Test-window IC of the V2d score by horizon: -0.020 at 1 month (t -0.87), +0.069 at 9 (t 1.98), +0.096 at 12 (t 2.43), +0.085 at 18 (t 2.00), +0.063 at 24 (t 1.86).	-0.02; -0.87; 0.069; 1.98; 0.096; 2.43; 0.085; 2.0; 0.063; 1.86	y
R2b	POST-HOC	The 12-month IC of the V2d score is +0.047 (t 0.90) in research and +0.071 (t 2.06) over the full sample.	0.047; 0.9; 0.071; 2.06	y
R3a	POST-HOC	Test Sharpe by holding period: +0.05 at 6 months, +0.14 at 9, +0.29 at 12, +0.30 at 18.	0.05; 0.14; 0.29; 0.3	y
R3b	POST-HOC	Research Sharpe by holding period: +0.31 at 6 months, +0.43 at 9, +0.50 at 12, +0.58 at 18.	0.31; 0.43; 0.5; 0.58	y
R4a	POST-HOC	V2d test Sharpe was -0.21 in the first half (71 months) and +0.64 in the second half.	-0.21; 71; 0.64	y
R4b	POST-HOC	V2d test Sharpe excluding Mar 2020-Dec 2021: +0.37; pre-2020: +0.14; 2020 onward: +0.37. Full-sample halves: +0.33 and +0.38.	0.37; 0.14; 0.37; 0.33; 0.38	y
R4c	POST-HOC	The 12-month IC of V2d was positive in both test halves (+0.097 and +0.095) even though the first-half Sharpe was negative.	0.097; 0.095	y
R5a	POST-HOC	V2d test Sharpe: +0.34 at 0 bp, +0.29 at 10, +0.22 at 25, +0.09 at 50 bp; break-even industry cost 68 bp (69 bp full sample).	0.34; 0.29; 0.22; 0.09; 68; 69	y
R6a	POST-HOC	With 12 HAC lags, V2d alpha is +0.26% a month in the test and +0.26% over the full sample.	0.26; 0.26	y
R6a2	POST-HOC	The t-statistics of the V2d alpha with 12 HAC lags are 1.48 (test) and 1.92 (full sample).	1.48; 1.92	y
R6b	POST-HOC	V2d loads against profitability in the test: RMW -0.36 (t -4.45) and SMB -0.17 (t -2.24); market, momentum and industry-momentum loadings are small.	-0.36; -4.45; -0.17; -2.24	y
R7a	POST-HOC	Dropping any one industry leaves the V2d test Sharpe between +0.18 (without health care) and +0.48 (without real estate), median +0.30.	0.18; 0.3; 0.48; Health care; Real estate	y
R8a	POST-HOC	V2d net Sharpe by regime at the decision (test): -0.00 when T rose over 12 months, +0.66 when it fell; +0.72 when V/U > 1, -0.14 when V/U < 1.	-0.0; 0.66; 0.72; -0.14	y
R8b	POST-HOC	The 12-month IC of V2d is highest when T is rising (+0.115, t 2.52) although the Sharpe in those months is zero.	0.115; 2.52	y
R9a	POST-HOC	Block-bootstrap 90% interval for the V2d Sharpe: [-0.07, +0.65] in the test (includes zero) and [+0.05, +0.64] over the full sample (excludes zero).	-0.07; 0.65; 0.05; 0.64	y
R11a	POST-HOC	Fundamental law, test window: A3 IC -0.021 x sqrt(156) x TC 0.80 implies IR -0.21 (realised -0.58); V2a IC +0.049 x sqrt(156) x TC 0.91 implies +0.56 (realised +0.16); V2d IC +0.096 x sqrt(13) x TC 0.48 implies +0.17, ceiling +0.35 (realised +0.29).	-0.021; 0.8; -0.21; -0.58; 0.049; 0.91; 0.56; 0.16; 0.096; 0.48; 0.17; 0.35; 0.29	y
R12	POST-HOC	With N_trials raised to 132, the V2d deflated Sharpe is 0.05 in the test and 0.16 over the full sample; the benchmark Sharpe that 132 null trials would produce is 0.76 (test) and 0.56 (full).	0.05; 0.16; 0.76; 0.56	y
R13	POST-HOC	Against the pre-declared battery rule, V2d fails only R4 (both test halves positive); R1, R2, R3, R5, R7 and R9 hold.	no; yes; yes; yes; yes; yes; yes	y

ID	Label	Claim	Numbers	Ver.
R10	POST-HOC	R10 (as-known ALFRED rerun) was not run: no FRED API key.	not run: no key	y

9 Known issues

- *V2c's ETF legs cancel.* Its long durable manufacturing and short professional services both map to XLI, so the tradable V2c book has four legs (book_2026Q4_etf.csv). Fix or disclosure: Disclose in the forward-test section; track V2c on French group returns as the primary record.
- *Revised vs as-known data.* Every historical v2/v3 number uses revised FRED values with fixed lags (option c in data_vintage.md). The frozen study found the same A3 IC under both rules, but as-known data picks the same top-3 groups in only 42% of months. Fix or disclosure: Label every v2/v3 number "revised data"; R10 (as-known rerun) is not run without a FRED key.
- *Overlap in the 12-month placebo.* The horizon-matched placebo for V2d uses overlapping 12-month returns; the circular shift preserves but does not correct the overlap, so its p-values are optimistic. Fix or disclosure: Keep the pre-declared 1-month placebo ($p = 0.85$) in the reading rule; report the 12-month one as a post-hoc check only.
- *Mapping dilution.* Labor data cover all establishments in an industry; returns cover listed firms. Durable manufacturing is 60% semiconductors and nondurable manufacturing 60% pharmaceuticals by market cap. Fix or disclosure: State as a limit; name NAICS-based CRSP baskets as the follow-up.
- *Power.* 142 test months detect an annualised Sharpe of about 0.6 at $t = 2$. None of the post-hoc Sharpe ratios (0.1-0.35) is distinguishable from zero after deflation. Fix or disclosure: Say so in the limits; the forward test is a record, not a verdict.